

UST'-KAMO, Russian Federation

WMO: 239920

Lat: 60.7298N

Lon: 97.5179E

Elev: 584

StdP: 14.39

Time Zone: 7.00 (E07)

Period: 86-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) -58.2	(c) -52.5	(d) -61.2	(e) 0.1	(f) -58.0	(g) -56.1	(h) 0.2	(i) -52.7	(j) 11.4	(k) -2.8	(l) 9.9	(m) -0.2	(n) 0.0	(o) N/A	(p) N/A

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a) 7	(b) 24.8	(c) 84.5	(d) 64.2	(e) 80.2	(f) 62.6	(g) 76.2	(h) 61.1	(i) 66.6	(j) 78.5	(k) 64.7	(l) 76.2	(m) 62.9	(n) 73.5	(o) 3.7	(p) N/A

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
62.7	87.0	70.0	60.7	80.9	68.6	58.6	75.0	67.0	31.7	78.3	30.2	76.2	28.8	73.5	75.2

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a) 13.5	(b) 11.1	(c) 9.2	(e) N/A	(f) N/A	(g) N/A	(h) N/A	(i) N/A	(j) N/A	(k) N/A	(l) N/A	(m) N/A	(n) N/A	(o) N/A	(p) N/A		
(a) 13.5	(b) 11.1	(c) 9.2	(e) N/A	(f) N/A	(g) N/A	(h) N/A	(i) N/A	(j) N/A	(k) N/A	(l) N/A	(m) N/A	(n) N/A	(o) N/A	(p) N/A		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	22.0	-24.2	-13.6	10.1	26.8	40.9	56.5	61.9	55.5	42.2	24.7	-1.9	-17.1
	DBStd	32.02	18.42	15.85	14.87	10.43	7.34	8.17	6.07	5.91	7.35	11.19	18.12	18.53
	HDD50	11040	2299	1781	1238	697	296	31	3	18	254	785	1557	2081
	HDD65	15749	2764	2201	1703	1147	746	274	130	297	684	1250	2007	2546
	CDD50	822	0	0	0	0	15	225	373	189	20	0	0	0
	CDD65	55	0	0	0	0	0	18	34	3	0	0	0	0
	CDH74	1210	0	0	0	0	10	505	604	89	2	0	0	0
	CDH80	310	0	0	0	0	1	147	158	4	0	0	0	0
(14) Wind	WSAvg	2.7	1.2	1.9	2.9	3.9	4.0	3.4	2.9	2.5	2.8	3.2	2.3	1.7
(15) Precipitation	PrecAvg	19.5	1.1	0.8	0.9	1.2	1.7	2.0	2.2	2.4	1.9	2.2	1.8	1.3
	PrecMax	24.4	2.2	1.6	1.8	1.9	3.0	3.4	3.8	3.9	3.3	3.4	3.5	2.2
	PrecMin	13.7	0.2	0.3	0.2	0.6	0.6	0.4	0.9	1.2	0.5	0.5	1.1	0.4
	PrecStd	2.5	0.4	0.3	0.4	0.4	0.6	0.7	0.8	0.7	0.6	0.7	0.6	0.5
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	23.0	28.8	43.8	58.5	75.5	89.9	88.1	80.2	72.1	50.8	37.6	29.9
		MCWB	22.2	26.2	35.1	43.6	55.7	65.9	66.8	63.7	58.0	42.8	35.9	29.0
	2%	DB	16.1	23.5	39.8	52.5	68.6	85.2	85.0	76.8	63.5	44.9	30.9	23.5
		MCWB	14.9	21.0	33.4	41.1	52.3	63.2	65.8	61.5	54.7	39.6	29.1	22.2
	5%	DB	10.0	18.7	35.7	47.3	63.2	80.5	81.1	73.5	58.5	41.2	27.4	17.3
		MCWB	9.0	17.1	30.8	37.8	48.6	61.7	63.6	60.2	52.3	38.0	25.8	16.2
	10%	DB	3.0	11.9	31.9	43.6	57.0	75.7	77.2	69.7	54.4	37.8	24.0	11.5
		MCWB	2.4	10.7	28.4	35.8	46.0	60.0	62.7	58.8	48.8	34.9	22.5	10.6
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	22.0	27.0	36.0	44.9	57.4	68.0	70.1	66.5	59.3	44.0	36.1	28.0
		MCDB	22.7	29.1	42.0	56.0	73.0	81.3	82.3	74.9	67.6	48.0	37.3	28.7
	2%	WB	15.1	21.5	33.6	41.5	53.8	65.4	67.7	64.0	56.2	40.5	29.4	21.8
		MCDB	16.2	23.2	39.0	50.7	64.7	80.4	79.2	72.8	61.2	44.0	30.7	23.0
	5%	WB	9.1	17.0	31.4	38.8	50.2	63.5	65.8	62.1	52.9	37.6	26.1	16.0
		MCDB	10.0	18.9	35.3	45.9	61.1	77.3	76.8	70.4	58.4	41.0	27.3	17.1
	10%	WB	2.5	10.7	28.6	36.5	46.8	61.1	63.9	60.0	49.6	35.2	22.5	10.6
		MCDB	3.2	12.0	32.1	42.5	56.0	73.7	74.4	67.5	53.5	37.2	24.3	11.5
(35) Mean Daily Temperature Range	5% DB	MDBR	17.3	26.6	30.4	25.6	22.6	26.2	24.8	22.5	18.1	15.3	19.1	17.8
		MCDBR	23.6	25.2	27.2	29.7	35.4	34.9	32.2	30.7	25.7	18.7	17.1	19.8
		MCWBR	22.6	23.4	21.3	19.0	19.6	15.7	15.1	17.0	16.1	13.7	16.1	19.1
	5% WB	MCDBR	23.4	24.8	25.9	25.6	29.0	29.9	28.3	25.0	23.4	18.3	16.8	19.9
		MCWBR	22.6	23.1	20.7	16.6	18.2	14.8	14.6	15.3	15.4	13.6	16.2	19.2
(40) Clear-Sky Solar Irradiance	taub		0.264	0.300	0.328	0.361	0.377	0.403	0.452	0.433	0.350	0.314	0.256	0.232
	taud		1.928	1.976	2.001	2.032	2.258	2.327	2.174	2.177	2.400	2.239	1.983	1.880
	Ebn at Noon		134	205	244	260	268	262	243	236	243	205	137	94
	Edh at Noon		15	27	37	44	38	36	41	38	25	21	14	10
(44) All-Sky Solar Radiation	RadAvg		134	429	892	1367	1572	1783	1614	1197	717	323	155	71
	RadStd		11	32	63	115	125	179	131	87	107	44	21	10

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (1 neighbor)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page